

The Application Layer: HTTP & The Web

How the Web Works: GET, POST, and Status Codes

IT 213 Instructor

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Where are we today?

Layer 7: Application (TODAY'S TOPIC) *(Data Unit: Message)*

Protocols: **HTTP (Web)**, SMTP (Mail), DNS

Content: User Data (HTML, Images, Text)

Layer 4: Transport *(Data Unit: Segment)*

Protocols: TCP, UDP

Header: Source/Dest **Ports**, Sequence Num, Checksum

Layer 3: Network *(Data Unit: Datagram)*

Protocols: IP (IPv4/IPv6), ICMP

Header: Source/Dest **IP Addresses**, TTL

Layer 2: Link / Physical *(Data Unit: Frame)*

Protocols: Ethernet, Wi-Fi, ARP

Header: Source/Dest **MAC Addresses**, Error Check

Source: Kurose & Ross (9th Ed): Figure 1.24 (Protocol Stack)

What is HTTP?

Definition

Hyper**T**ext **T**ransfer **P**rotocol is the language web browsers and servers use to talk to each other.

How it works (The Conversation):

- **Client (Chrome):** "I want to see /index.html."
- **Server (Apache):** "Here is the file." (or "I can't find it").

Key Concept: Stateless

- The server forgets you the moment it sends the file.
- It does NOT remember if you visited 5 seconds ago. (That's why we need Cookies!)

Source: Kurose & Ross: Section 2.2.1 (Overview of HTTP)

Request Method 1: GET

GET = "Give me something"

Used when you want to **retrieve** data (load a page, view an image).

Characteristics:

- Data is visible in the URL (e.g., search?q=cat).
- **NOT** secure for passwords.
- Can be bookmarked.

```
GET /index.html HTTP/1.1
Host: www.concordia.ab.ca
User-Agent: Mozilla/5.0 (Windows 10)
```

Request Method 2: POST

POST = "Take this"

Used when you want to **send** data to the server (Login, Upload, Form Submit).

Characteristics:

- Data is hidden inside the "Body" of the message.
- **Secure** (if using HTTPS).
- Cannot be bookmarked.

```
POST /login.php HTTP/1.1
Host: www.concordia.ab.ca

username=student&password=12345
```

The Server's Reply: Status Codes

When the server answers, it gives a 3-digit code to say what happened.

Range	Meaning	Famous Examples
200-299	Success	200 OK (Here is your file)
300-399	Moved	301 Moved Permanently (Go here instead)
400-499	User Error	404 Not Found (You typed it wrong)
		403 Forbidden (No permission)
500-599	Server Error	500 Internal Error (My code crashed)

Source: Kurose & Ross: Section 2.2.3 (HTTP Message Format)

Summary: What you need for Lab 1

- ❶ **Layer 7 (Application):** Where the browser lives.
- ❷ **HTTP:** The text-based protocol we use.
- ❸ **GET vs POST:**
 - **GET:** Reading (Safe, URL visible).
 - **POST:** Writing (Forms, Hidden body).
- ❹ **200 vs 404:**
 - **200:** Good.
 - **404:** Bad (File missing).

Next Up: We will use Wireshark to capture these "GET" packets!